

Aditya Dutta

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SUMMARY

Third-year undergraduate building electromechanical hardware end to end: mechanical design and fabrication (3D printing, CNC, laser cutting, fixtures and jigs), PCB design, firmware and closed-loop motor control. Technical lead of a funded startup shipping a 100-unit batch; author of an open-source CAN-FD field-oriented motor controller. Seeking a robotics control or product design internship.

EDUCATION

Manipal Institute of Technology, Bengaluru | B.Tech, Computer Science and Engineering; Minor in Finance (TAPMI) Expected 2028

TECHNICAL SKILLS

Control & Robotics: Field-oriented control (Park/Clarke, SVPWM), PID current / velocity / position loops, state estimation and IMU sensor fusion, magnetic encoders, real-time motor control on STM32, Jetson + MCU architectures, CAN-FD actuator networks, ROS 2

Mechanical Design & Fabrication: Fusion 360 CAD, mechanism and actuator packaging, enclosure and gasket design, tolerance stack-up, DFM/DFA, fixture and jig design, rapid prototyping, FDM 3D printing, CNC machining, laser cutting, brazing, manual PCBA

Embedded & Firmware: Embedded C/C++, Python, MATLAB, STM32 (G4, F4), ESP32-C3, bare-metal real-time firmware, low-power design, SPI, I2C, UART, CAN-FD, SWD, Linux, Git, oscilloscope and logic-analyser bring-up

Power Electronics & PCB: Multi-layer PCB design (Altium), three-phase BLDC gate drive and MOSFET selection, Li-ion charging and battery management, switching regulators and LDO rails, current sensing, thermal design, PCB DFM

EXPERIENCE

Co-Founder & Technical Lead | Sylvara Inc., Ontario, Canada (remote) Apr 2025 - Present

- Designed a 22 x 42 mm multi-layer IoT plant-care board (ESP32-C3-WROOM-02, TI BQ25798 solar harvesting, USB-C PD) at 53 microamps average draw - a calculated 1.3 years without sun; released v1.3 to manufacture as a 100-unit first batch.
- Owned schematic, layout, firmware, enclosure fit, BOM and DFM through four revisions: TLC555 capacitive front end replacing noisy GPIO sensing, 0402 to 1206 passives, USB ESD/TVS protection, IPX9 ingress-protection gasket and board-to-enclosure tolerancing.

President | RADAR Student Robotics Club, MIT Bengaluru 2025 - Present

- Set technical direction for the TrackMate autonomous track pacer (with Tenvik Sports), Arachne legged walker and combat robotics teams.
- Co-designed the club's Arduino/NRF24 RC teaching platform on a compact interconnection PCB, used to teach drivetrain and motor control.

Student Researcher | MIT-B with ISRO and the Indian Air Force 2026 - Present

- Building the sensing chain and embedded integration for a mmWave-radar mobile runway foreign-object-debris (FOD) early-warning system.

PROJECTS

FlexNode | open-source distributed CAN-FD motor controller, Apache-2.0 2026

- One CAN-FD node fusing three-phase field-oriented drive with a sensor-fusion front end: STM32G473CEU6 at 170 MHz, TI DRV8353 into six 60 V / 1.6 mOhm MOSFETs, 10-44 V input, 100 A peak phase, 900 W at 30 V, 15-30 kHz current loop.
- Forked mjbots moteus r4.11, keeping the power stage; packaged an LSM6DS3TR-C IMU, optional VL53L7CX ranging, servo/aux and I2C ports and a 5 V / 3 A buck onto the per-actuator board, with a capability-discovery register block so one firmware image serves the fleet.
- Brought up first boards and spun a motor open-loop within 48 hours behind bus-sag, overcurrent and FET-temperature aborts; corrected a phase-swap layout error in the firmware hardware layer instead of respinning.

CatBot | biomimetic quadruped, 19 + 2 DoF, 4.8 kg 2025 - Present

- Architected a decentralised drivetrain - one FlexNode per actuator closing a 15-30 kHz current loop at the joint over a 5 Mbit CAN-FD spine split left/right for fault isolation; 2.5 kW peak from a 266 Wh 8S2P Molicel pack.
- Jetson Orin Nano Super running stereo perception and visual-inertial SLAM above an STM32 reflex-and-watchdog brainstem.
- Relocated six knee motors from the distal limb onto 14:1 dual-stage belt reductions to cut swing-leg inertia; modelled the pulley stack, belt path and motor packaging in Fusion 360.

RoVolt | WHEG rover and mobile agricultural platform 2023

- Designed and fabricated the rover end to end - laser-cut, hand-brazed 6061 aluminium chassis, 3D-printed wheel-leg (WHEG) hubs on push-rod suspension, Raspberry Pi 3 with TB6612FNG drivers and quadrature encoders - then reconfigured it into a three-wheel agricultural platform with a soil-moisture, ploughing and seeding arm and IMU heading.

AWARDS & RECOGNITION

- 1st place, Smart Bengal Hackathon (RoVolt, 2023); CAD 10,000 seed funding and Morrisette Institute backing for Sylvara Inc. (2025).